

1. Genetic Algorithm

Given:

- Objective function:

$$f(x) = x^2$$

- Initial population:

$$\{2, 4, 6, 8\}$$

Step 1: Calculate Fitness Values

The fitness of each individual is calculated using:

$$f(x) = x^2$$

Individual (x) Fitness $f(x) = x^2$

$$2 \qquad 2^2 = 4$$

$$4 \qquad 4^2 = 16$$

$$6 \qquad 6^2 = 36$$

$$8 \qquad 8^2 = 64$$

Fitness Scores

- $2 \rightarrow 4$
- $4 \rightarrow 16$
- $6 \rightarrow 36$
- $8 \rightarrow 64$

Step 2: Selection

Individuals with the highest fitness are selected.

Selected candidates:

- **8 (Fitness = 64)**
- **6 (Fitness = 36)**

These are the best individuals because they have the highest fitness values.

Step 3: Crossover

Convert the selected individuals into binary.

- $8 = 1000$
- $6 = 0110$

Exchange the **Least Significant Bit (LSB)**.

Before crossover:

- Parent 1 = **1000**
- Parent 2 = **0110**

LSBs:

- Parent 1 LSB = **0**
- Parent 2 LSB = **0**

After exchanging:

- Offspring 1 = **1000** (8)
- Offspring 2 = **0110** (6)

Since both LSBs are the same, the offspring remain unchanged.

Generated Offspring:

- Offspring 1 = **8**
- Offspring 2 = **6**

Step 4: Role of Mutation

Mutation randomly changes one or more bits of an individual with a small probability.

Importance of Mutation

- Increases genetic diversity.
- Prevents premature convergence.
- Helps escape local optima.
- Explores new possible solutions.
- Improves the chances of finding the global optimum.

Example:

If offspring **1000 (8)** undergoes mutation by flipping the last bit:

$$1000 \rightarrow 1001$$

New value:

$$1001_2 = 9$$

Fitness becomes:

$$9^2 = 81$$

Thus, mutation may generate an even better solution.

Final Answer

Fitness values:

- $2 \rightarrow 4$
- $4 \rightarrow 16$
- $6 \rightarrow 36$
- $8 \rightarrow 64$

Selected individuals:

- **8 and 6**

Offspring after LSB crossover:

- **8 and 6**

Mutation improves diversity, prevents stagnation, and helps discover better solutions.

2. Perceptron Model

Given

Weights:

$$w_1 = 0.5, w_2 = 0.3$$

Bias:

$$b = -0.4$$

Inputs:

- Advertisement exposure:

$$x_1 = 2$$

- Income level:

$$x_2 = 1$$

Step 1: Calculate Net Input

The perceptron net input is

$$\text{Net} = w_1x_1 + w_2x_2 + b$$

Substitute the values:

$$\begin{aligned} &= (0.5 \times 2) + (0.3 \times 1) + (-0.4) \\ &= 1 + 0.3 - 0.4 \\ &= 0.9 \end{aligned}$$

Therefore,

$$\boxed{\text{Net Input} = 0.9}$$

Step 2: Apply Step Activation Function

Step activation function:

$$y = \begin{cases} 1, & \text{if Net} \geq 0 \\ 0, & \text{if Net} < 0 \end{cases}$$

Since

$$0.9 > 0$$

Output:

$$\boxed{y = 1}$$

Step 3: Interpretation

Output = **1**

Therefore, the perceptron predicts that **the customer will purchase the product.**

Final Answer

Net input:

$$\boxed{0.9}$$

Activation output:

$$\boxed{1}$$

Decision:

The customer makes a purchase.